Raising time-F1 and content-F1 on OmniPro Online

A diagnosis of the Qwen2.5-Omni fork, and what to do next

Threshold-fit working notes

1 September 2026

The short answer. The audio/video interleaving is *not* the problem — three separate checks clear it. The problem is upstream of the gate: the ICL control DSL that works on Qwen3-VL-8B **does not transfer** to Qwen2.5-Omni-7B. The controller generates a mean of **21.7 tokens per tick against a 300-token budget**, stopping on its own before the schema is finished; `fps` appears in **12 of 78,828** ticks and `next_check_s` in **8**. The gate is thresholding a boolean produced by a generation step that is not working, which is why AUC ≈ 0.5 and why no threshold helped. That is a *fixable upstream cause*, not a property of the task.

1 Where we actually are

Stage 3, arm 1 (fitted per-task thresholds), 2,512 of 2,700 samples banked, judge offline so content metrics are WITHHELD.

stratum	n	n_{gt}	n_{emit}	t-P / t-R	**time-F1**
audio required	1643	4916	16185	0.155 / 0.510	0.2375
audio not required	868	3511	12866	0.164 / 0.601	0.2578
gross	2511	8427	29051	0.159 / 0.548	**0.2464**

Two facts frame everything below.

(1) The system emits 3.4× more often than there are events (29,051 emissions against 8,427 ground-truth triggers). Precision, not recall, is the binding constraint.

(2) The gate is nearly exhausted. §2.5 measured precision as *flat* across the whole threshold grid. With p pinned at 0.159, $F_1 = 2pr/(p+r)$ climbs monotonically as $r \to 1$ to a ceiling of $2p/(p{+}1) = \mathbf{0.274}$. We are at 0.246. Every remaining gate adjustment in existence is worth ≤ 0.028 — *inside* the 0.03 noise band — and collecting it means emitting even more. Note the corollary, which is counter-intuitive: cutting the 3.4× over-emission would *lower* F_1, because precision will not rise to compensate.

1.1 Two different failure modes, not one

	task	gt/vid	emit/vid	time-F1
A. over-emitting	sequential_step_instruction	5.65	17.24	0.3778
	realtime_state_monitor	4.43	22.43	0.2431
	cumulative_counting	4.76	11.23	0.2337
	event_narration	4.37	19.22	0.2285
	dedup_counting	4.04	16.07	0.2097
	semantic_condition_alert	3.26	14.30	0.2071
B. right count, wrong moment	explicit_target_grounding	1.10	1.05	0.1065
	snapshot_counting	1.00	1.00	0.0674
	instant_event_alert	1.31	1.00	0.0604

`snapshot_counting` isolates the timing problem. Exactly 1.00 ground-truth event per video, exactly 1.00 emission per video, and time-F1 0.0674 — so the single emission lands within ±3 s only about 7% of the time. Per-emission timing, not emission volume, is what fails here.

But volume is *also* a lever on these tasks, and §5's comparison proves it. The parent system scores 0.2776 on `snapshot_counting` while emitting **2.85** times per ground-truth event, against our 1.00. Inverting $F_1 = 2\,tp/(k+1)$ for one GT event and k emissions: the parent converts ≈53% of videos into a hit with 2.85 attempts (≈24% per attempt); we convert 6.7% with one. *Both* factors are real — our per-emission timing is roughly 3.5× worse *and* we take a third as many attempts. The 600 s / 300 s refractories inherited from the vision-only fit are therefore binding on all three group-B tasks, not just `instant_event_alert`.

2 Five hypotheses, tested

hypothesis	verdict	evidence
TMRoPE positions are wrong	× cleared	numerically validated against `get_rope_index` (FORK §3); a 2 s / 50-token audio chunk from anchor 8 yields exactly 8 . . . 57
Position clock outruns trained RoPE range	× cleared	max position reached **≈10,300** vs `max_position_embeddings` =32,768; the loud guard fired **0** times in 2,500+ videos; eviction never fired
Frozen perception (the parent's defect #1)	× cleared	median **110 distinct `seen`** per video over 187 ticks; **0 of 532** videos emitted a single frozen description
Audio is not actually reaching the model	× cleared	`use_audio=True` on **2,514/2,514**, including all 1,645 `audio_dependency=required`
The ICL DSL does not transfer	✓ **confirmed**	§3

3 What is actually broken

3.1 The controller stops generating after ∼22 tokens

measurement (78,828 controller ticks, 4 lanes)	value
`controller_max_tokens` budget	300
mean tokens actually generated per tick	**21.7** (max 108)
ticks emitting `seen`	100%
ticks emitting `answer`	51%
ticks emitting `event_time_s`	41%
ticks emitting `fps`	**12** (0.015%)
ticks emitting `next_check_s`	**8** (0.010%)
ticks emitting `question_for_next`	21 (0.027%)
probe argmax is literally `true/false`	84.5%
probe argmax agrees with the parsed JSON	77%

The budget is 300 and the model uses 22. It is not being truncated — it **chooses to stop**, mid-schema. Representative `seen` payloads from the live logs:

`"1 prices are ...' \n * Do not other numbers"` • `"{a single CJK character}"` • `"0\n{"`

with `answer` fields reading `"{"` and `"1.0,"`. The JSON parses — 0% unparseable — so nothing raises an error. This is the project's signature failure mode: *a finished run and a number you could put in a table.*

Why this explains AUC ≈ 0.5. `have_enough_info` is the boolean the gate thresholds. It is produced by the same degenerate generation step. A threshold on a boolean that the model is not really computing cannot rank event-adjacent ticks above quiet ones — so the four independent measurements of "no threshold helps" were all measuring one upstream defect, not nine hard tasks.

3.2 The self-pacing loop is dead

`fps` and `next_check_s` are how the ICL controller was supposed to *schedule itself* — look more often when something is happening, less when it is not. They appear in 20 ticks out of 78,828. The system is running at a fixed cadence with its adaptive machinery present in the prompt and absent from the output. This is the parent branch's defect #7 ("fields simply never appeared") recurring in a new form.

3.3 A verified structural divergence: the modality delimiters

The reference processor builds an interleaved A/V stream as

`<vision_bos><audio_bos> [video chunk` j`][audio chunk` j`]`* `<audio_eos><vision_eos>`

Our *ordering* matches it: video frames for $[t, t{+}2)$, then the 2 s audio chunk for the same window. But **none of the four delimiter tokens appears anywhere in `async_omni_v2/`**. The model is fed an interleaved stream with no marker saying an interleaved stream has begun.

This is a real divergence from training conditions, verified from the reference source. It is **not** established as a *cause* — no ablation has been run. It is cheap to test and belongs high on the list precisely because it is cheap, not because it is proven.

4 Why the original ICL branch appears to score so much higher

The number is `joint_f1` **0.206**, recorded in `EXPERIENCE.md` as "roughly MiniCPM-o-4.5-level". Its own parenthetical is the important part: *"27-video, 9-task eval, older scoring."*

`LEARNINGS.md` §6b then documents what "older scoring" meant — six defects, all in the scorer:

#	defect	effect
13	LLM judge silently fell back to **word overlap** on any exception, while logging "google-genai judge active"	every j
17	state matched by **substring**, not equality	"RSM
15	`semantic_condition_alert` LLM-judged; upstream scores it `time_only`	wrong
14	`judge_cache.json` clobbered by 4 concurrent writers	24 h of
16	`_extract_count` did not strip clock times: "At 01:23, count is 5" = 1	correct

Defect #13 was caught by re-scoring in a bare shell **with no API key at all**, which reproduced the shipped `content_acc` *to four decimals*. A real judge and no judge cannot agree exactly — so no judge had ever run.

Aligning to the benchmark's real scorer then found `realtime_state_monitor` at **0.000, 296 of 296 wrong**: it emitted sentences where the benchmark compares the whole payload to `state_to` by equality. That fix (emit the bare label) is commit `ab49ce9`, and it *is* present in this fork's prompts.

And the deepest one, `LEARNINGS` §2: a null model using **only the timestamp**, with zero pixel access, beat `p_hit` on 3 of 4 tasks; `p_hit` correlated with video time at $|r|$ up to **0.93**. In the repo's own words: *"The system was scoring worse than a clock."*

Conclusion. 27 videos against our 2,700, a judge that was never running, and a control signal that was largely a clock. The 0.206 is not a target to chase. What *is* real in that branch are its **mechanisms**: hysteresis with re-arm (`hyst2b`), the per-tick `seen` step, word-overlap dedup within a true stretch, and the "already reported" history. Those survive the audit; the score does not.

5 Head to head: the parent's fitted table vs ours

The parent's own fitted per-task table (`output_full9/fitted_full_table.json`, Qwen3-VL-8B, vision only, its shipped thresholds, **judge running**) against stage-3 arm 1 (Qwen2.5-Omni-7B, vision + audio, fitted thresholds, judge off). **time-F1 only** — it is the one metric both sides measure the same way.

	system_3 Qwen3-VL-8B (V)			**omni** Qwen2.5-Omni (A+V)			
task	n	e/gt	time-F1	n	e/gt	time-F1	Δ
cumulative_counting	101	3.50	0.2466	283	2.36	0.2337	−0.013
dedup_counting	283	0.67	0.3059	278	3.95	0.2097	−0.096
event_narration	96	2.56	0.1697	279	4.40	0.2285	**+0.059**
explicit_target_grounding	12	3.07	0.2295	280	0.95	0.1065	−0.123
instant_event_alert	38	1.33	0.2105	272	0.76	0.0604	−0.150
realtime_state_monitor	142	1.06	0.2138	268	5.06	0.2431	**+0.029**
semantic_condition_alert	38	1.42	0.2644	286	4.39	0.2071	−0.057
sequential_step_instruction	148	1.08	0.3905	298	3.05	0.3778	−0.013
snapshot_counting	73	2.85	0.2776	267	1.00	0.0674	**−0.210**
overall (micro)	931	1.48	**0.2684**	2511	3.45	**0.2464**	−0.022
macro over tasks			**0.2565**			**0.1927**	**−0.064**
t-P / t-R		0.225 / 0.333			0.159 / 0.548		

The omni fork wins on 2 of 9 tasks. Micro-averaged it trails by 0.022 — inside the 0.03 noise band — but **macro-averaged it trails by 0.064**, which is not. The micro figure flatters us because it is dominated by the tasks we do adequately; the macro figure exposes that our three worst tasks are much worse.

The two systems fail in opposite directions. The parent emits **1.48** times per event at precision 0.225 / recall 0.333; we emit **3.45** times at precision 0.159 / recall 0.548. It is conservative and accurate; we are talkative and vague. Our advantage on `event_narration` and `realtime_state_monitor` — both `level`-mode, short-refractory tasks — is exactly where talking more pays.

Where we lose most, the parent simply gets more attempts. On the three group-B tasks the parent emits 2.85 / 3.07 / 1.33 times per event where our inherited refractories permit 1.00 / 0.95 / 0.76.

Read this table with four caveats. (i) *Different subsets* — the parent's n is uneven and small on exactly the tasks where it wins most (`explicit_target_grounding` n=12, `instant_event_alert` n=38); ours is ≈280 per task. Those two rows are the weakest in the table. (ii) *Different modality and backbone* — this is not an ablation of audio, it is two different systems. (iii) *Content is not comparable at all*: the parent ran with a working judge (`n_unjudged`=0), ours is WITHHELD. Only time-F1 is being compared, and only because the ±3 s greedy match is the same code path in both trees. (iv) The parent's numbers are read from another account's scratch, which this project has already been burned by once; they should be re-derived before publication.

6 What to do, in priority order

Ordered by (evidence it will help) × (cheapness), not by ambition.

Tier 1 — attacks the confirmed root cause

1. **Constrain the decode to the schema.** The model stops mid-JSON at 22 tokens. Guided/constrained decoding (a grammar or a logit-processor mask over the schema) makes the whole DSL structurally unskippable. This alone restores `answer` on the 49% of ticks that currently lack it and `event_time_s` on the 59% that lack it. *Cost: hours. Expected effect: large and broad.*

2. **Shrink the schema for this model.** Qwen2.5-Omni-7B is a weaker structured generator than Qwen3-VL-8B. Two fields (`have_enough_info`, `event_time_s`) may outperform seven that it never finishes. Test the reduced schema against the full one at equal budget. *Cost: hours.*

3. **Use `event_time_s` — the gate currently ignores it.** The model already names an onset time on 41% of ticks, and the emission is timestamped at the *crossing tick* instead. Emitting at the model's stated time is a different, possibly much better, timing signal that costs **no GPU** to evaluate: it is already in the banked logs. *Do this first — it is free.*

Tier 2 — cheap, plausible, unproven

4. **Add the four modality delimiters** (§3.3) and A/B one task. Cheap, and it restores a training-time invariant we are currently violating.

5. **Detrend `p_hit` within each video.** The parent measured $|r|$ up to 0.93 between `p_hit` and wall-clock. If that holds here, most of the score is a clock; z-scoring or differencing within a video could expose the residual. **Free** — offline, from banked tick logs.

6. **Gate on the derivative, not the level.** Events are *changes*. A rising-edge gate on a level signal fires wherever the level is high. Also free to test offline.

Tier 3 — known-bounded gains

7. **Refit mode and refractory** (never refitted; inherited from the vision-only model). *Promoted after the §5 comparison*: the parent emits 2.85× per event on `snapshot_counting` and 3.07× on `explicit_target_grounding` where our 600 s / 300 s refractories permit 1.00, and it scores 0.28 and 0.23 against our 0.07 and 0.11. This is worth more than the earlier ceiling analysis suggested — it binds on all three group-B tasks, which are our three worst.

8. **Do not chase the emission budget.** Matching emissions to event count would *reduce* F_1 while precision stays flat. Revisit only after Tier 1 makes the score informative.

Content-F1 specifically

9. **Restore a judge.** `content_acc` and `joint_f1` are WITHHELD — never guessed — and will stay so until one is reachable. The true value lies between the lower bound 0.038 and the timing ceiling 0.246; nothing narrows that without a judge. This blocks any comparison with MiniCPM-o 4.5 (0.209 *joint*-F1, not time-F1, and judged by a different model on a different subset).

10. **Fix `answer` degeneracy** — same root cause as item 1. An `answer` of `"{"` scores zero content no matter how good the timing is, so *every* joint-F1 point is currently gated behind Tier 1.

11. **Emit the benchmark's token form, not prose**, per task. Already done for `realtime_state_monitor`; audit the remaining constrained-format tasks the same way.

7 The honest bottom line

The threshold-fit study's negative result stands and is well-evidenced: *given this control signal*, no per-task threshold recovers information the signal does not carry, and the fit does not overfit (fit-disjoint bias +0.0007 gross). What the study could not see from inside is that **the signal itself is the artefact of a broken generation step**. The right next move is not a better gate. It is to make the controller finish its sentence.

Two caveats on this document. (i) The DSL-collapse diagnosis is measured; the claim that fixing it raises F_1 is a *hypothesis*, and item 3 is the free experiment that tests it before any GPU is spent. (ii) The audit of the parent branch's 0.206 is read from this repository's own post-mortems; that branch has not been re-run here.